When should a neural text-entry decoder commit? A decoder-agnostic, risk-controlled trust layer evaluated on intracortical, EMG and MEG data

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Abstract

Objective. Neural text-entry decoders increasingly end in a language model that turns weak neural evidence into fluent sentences, but they rarely signal when a sentence is wrong. We ask when such a system should commit a sentence, offer alternatives or abstain, and whether one decision rule can serve decoders for different recording modalities. *Approach.* We define an open per-frame output format (NeuralTextStream) and a trust layer that reads only this stream. The layer calibrates confidence, chooses between commit, offer and abstain, and sets its commit threshold with conformal risk control (an expected-risk bound under exchangeability), Learn-then-Test (a high-probability bound) or adaptive online updates from user corrections (a long-run bound). We re-fit the same implementation offline on public data from one intracortical speech participant (816 test trials), forearm surface EMG key presses composed into sentences (19 participants) and MEG typing (19 participants). A simulated eye-gaze user scores each policy in effective words per minute (eWPM) against always committing, always offering and gaze typing alone. *Main results.* On intracortical speech, the layer cut wrong commits from 61% to 6% of sentences at the speed of always committing (11.55 vs 11.45 eWPM) and 1.6% faster than always offering, but one committed sentence in four was still wrong. As a gate on the published decoder, it cut wrong commits from 19% to 4% at a 4.3% speed cost. On EMG it committed at most 4% of sentences. On MEG it never committed, and speed did not differ detectably from gaze typing. Conformal calibration held the mean wrong-commit rate at 0.048 (target 0.05) over random splits; online re-calibration kept it at 0.040–0.053 where a threshold fixed on earlier data reached 0.0625–0.0635 **[TODO: R1: re-check after the clean-calibration drift baseline]**. The layer committed no text from signal-destroying control streams, but it cannot detect a real stream of a different intended sentence. *Significance.* A stream-level commit rule with stated guarantees can make language-model-assisted neural text entry safer without slowing it when the decoder is accurate enough. All speed results are simulations and need a user study.

1 Introduction

Brain–computer interfaces (BCIs) for text entry are moving from the laboratory towards clinical and consumer use. Intracortical speech neuroprostheses decode attempted speech at conversational rates [1, 2]; handwriting and typing neuroprostheses reach similar speeds [3, 4]; non-invasive decoders reconstruct typed sentences from magnetoencephalography (MEG) [5, 6]; and wrist or forearm surface electromyography (sEMG) is a candidate input for wearable devices [7–9]. Most of these systems end in a language-model (LM) stage that turns noisy per-frame predictions into sentences. This stage lowers the error rate, but it changes the kind of error: when the neural evidence is weak, the LM still returns a grammatical sentence. In speech recognition, fluent output that the input does not support has caused documented harm [10], because a reader has no cue that it is wrong. For a person who depends on the interface to communicate, a wrong sentence that is sent in their name can cost more than a slow one.

BCI spellers have long handled this problem inside one system. P300 and rapid serial visual presentation (RSVP) spellers stop collecting evidence when the posterior of a character passes a threshold [11–13], often with an LM prior [14–16]; error-related potentials can flag a wrong selection after the fact [17, 18]; and myoelectric pattern recognition rejects low-confidence decisions [19]. Each of these rules is built into one decoder and one paradigm. Sentence-level LM decoders for speech, handwriting, typing and sEMG now share a common structure: per-frame probabilities followed by an LM search. This shared structure makes a decoder-independent commit rule possible. Such a rule chooses between committing a sentence, offering alternatives and abstaining, comes with a finite-sample bound on how often it commits wrong text, and works on any decoder that exposes per-frame probabilities.

Three things are missing for such a rule. First, there is no common output format: the Apple BCI Human Interface Device profile sends discrete events with a 0–255 "neural activity strength" that is not a probability [20], W3C EMMA has an n-best list but no per-frame distributions [21], and Lab Streaming Layer and BIDS carry raw signals, not decoded text [22, 23]. Second, calibration [24], selective prediction [25] and conformal risk control [26] are well studied, and recent work adds calibrated confidence or alternatives to single BCI decoders [27, 28] and risk control to LM correction in speech recognition [29]. To our knowledge, no work has combined these into a commit rule for neural text entry, with a stated guarantee, and tested it on more than one recording modality. Third, error rates do not show whether a BCI saves the user time. A no-BCI baseline, such as gaze typing [30], and an "always offer, never commit" baseline, as used by most spellers, are rarely reported next to the decoder.

This work. We define the problem (Section 3), a minimal per-frame output format (Section 4), a trust layer that reads only this format (Section 5) and an evaluation protocol (Section 6). We test them offline on public data from three recording modalities. Our contributions are:

1. **A decoder-agnostic trust layer** that reads only per-frame log-probabilities: temperature calibration, a neural-evidence guard, and a COMMIT/OFFER/ABSTAIN policy fitted for maximum eWPM under a wrong-commit limit. The same implementation is re-fitted per device (Table 1). It also gates the published sentence of an LM-rescored decoder whose weights we do not have, with the help of a light reference decoder that runs on the same stream.
2. **Wrong-commit control with stated guarantees.** We set the commit threshold with conformal risk control (expected-risk bound under exchangeability) or Learn-then-Test (high-probability bound) [26, 31]. We re-tune it online from user corrections with adaptive conformal updates, which bound the long-run average of the observed wrong-commit rate for any sequence of sentences [32, 33].
3. **An evaluation protocol for commit decisions**: eWPM of a simulated user against gaze typing, always-commit and always-offer baselines; the wrong-commit rate together with the commit rate and the error rate among commits; and two classes of controls, signal-destroying streams and real streams of a different intended sentence.
4. **An open format and release**: the NeuralTextStream v0.1 specification (per-frame log-probabilities with timing, calibration and provenance conventions), adapters for MEG, intracortical and sEMG decoders, a hash-chained audit log, and all experiment scripts under Apache-2.0 (Section 12).

Figure 1 summarises the main result. The commit decision was tested non-trivially on one intracortical participant: there the layer turned frequent wrong commits into rare ones at the speed of always committing, and it was slightly faster than always offering. On sEMG, with sentences composed from isolated key presses, it committed at most 4% of sentences, so it behaved almost like an always-offer policy. On MEG, where the decoder almost never ranked the right sentence first, the layer correctly never committed. MEG is therefore a negative control for the commit decision, not a third positive case.

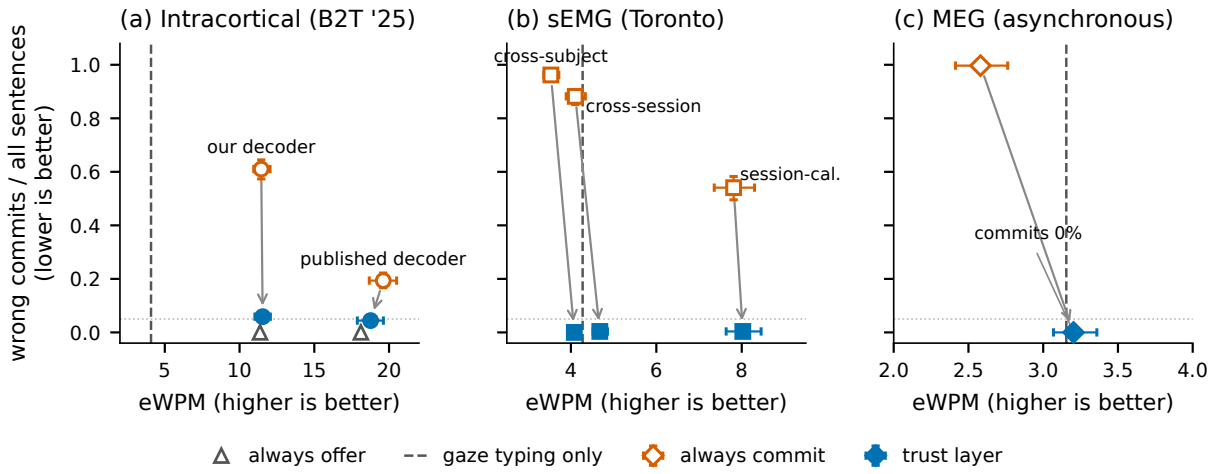


Figure 1: The same trust layer turns frequent wrong commits into rare ones on three device types; on MEG it does so by never committing. Each panel plots wrong commits per sentence attempted (wrong commits / all sentences) against effective words per minute (eWPM) of a simulated user. Open markers: always commit the decoder's top sentence. Filled markers: the trust layer on the same decoder output; arrows connect the two. Triangles: always offer the top candidates and never commit (B2T '25; point values). Dashed line: gaze typing alone (no BCI). Dotted line: target $\alpha = 0.05$. Error bars: 95% bootstrap CIs over trials (a) and sentences (b, c; cluster bootstrap over the 17 unique sentences in c). (a) Intracortical speech, Brain-to-Text '25, one participant, 816 test trials; "our decoder" is our phoneme-to-word decoder, "published decoder" is the published LM-rescored sentence. (b) Forearm sEMG, 551 composed test sentences from 19 participants, three decoder conditions. (c) MEG typing, 637 test instances of 17 unique sentences from 19 participants, our asynchronous decoder; here the trust layer offered candidates on every instance, so it is the always-offer policy. **[TODO: R1: add always-offer points for sEMG from the always-offer analysis.]**

2 Related work

Brain-to-text decoders. Non-invasive MEG decoding of typed sentences reached a mean character error rate (CER) of 0.29 with a keystroke-locked (synchronous) model [5]. A second version decodes sentences from the continuous recording without key times (asynchronous) and adds an LLM stage [6]. Other non-invasive work decodes perceived speech [34] or words with large pre-trained models [35, 36]. Intracortical decoders reach high accuracy for attempted handwriting [3], speech [1, 2, 37] and typing [4]; public benchmarks expose their phoneme logits [38, 39]. Surface EMG text entry now exists at scale for the wrist [7, 9] and the forearm [8]. Most of these decoders use connectionist temporal classification (CTC) or a related sequence loss [40] with beam search and an LM [41]. In the intracortical speech study whose data we use, decoded words were displayed on a screen and completed sentences were then read aloud by text-to-speech [2]. Our layer acts at this point: it decides whether a decoded sentence is committed (sent or spoken), offered for confirmation, or not shown. **[TODO: R1: check in the supplement of Card et al. whether the participant confirmed each sentence before text-to-speech; describe that interaction if so.]**

Error handling in BCI spellers. Spellers based on P300 or RSVP commit a character when its posterior passes a threshold, and stop collecting evidence early when confidence is high [11, 12]; with users with ALS, dynamic stopping raised communication rates [13]. LM priors are fused with the EEG evidence in these spellers [14–16]. Error-related potentials (ErrPs) in the EEG after a wrong selection allow automatic error detection and correction [17, 18, 42–44]. They are the BCI field's answer to errors that the user sees but does not correct explicitly, and a possible implicit feedback signal for our online controller. Asynchronous BCIs detect a "no-control" state, so that no output is produced when the user does not intend to communicate [45–47]. In

myoelectric pattern recognition, confidence-based rejection withholds uncertain decisions [19]. These rules are commit/abstain rules with an accuracy target, built into one decoder and one paradigm. Our rule acts on sentence-level LM output, reads only the decoder's per-frame output, and states a finite-sample bound.

Adaptive decoders. Closed-loop decoder adaptation [48], self-calibration during virtual typing [49], manifold alignment [50] and unsupervised recalibration [51, 52] keep intracortical decoders accurate over days. These methods change the decoder. Our online controller leaves the decoder fixed and re-tunes only the commit threshold from user feedback; the two approaches can be combined.

Calibration, selective prediction and abstention. Modern neural networks are overconfident; temperature scaling is a simple and strong fix [24], but calibration degrades under dataset shift [53]. The maximum softmax probability is a strong baseline for detecting errors [54], and selective classification trades coverage for risk [25]. In BCI, confidence-aware decoding with abstention was shown for EEG speech commands on one device [27], and better calibration and alternatives improved the fusion of brain-to-text output with other inputs [28]. Spoken dialogue systems accept or reject a recognised utterance by comparing its confidence with thresholds set from the costs of misunderstandings and false rejections [55]. This is a commit/abstain decision under another name, without a finite-sample guarantee.

Conformal prediction and risk control. Conformal methods give distribution-free, finite-sample guarantees [56]. Conformal risk control (CRC) bounds the expected value of a monotone loss [26]; Learn-then-Test (LTT) bounds the risk with high probability by multiple testing [31]. Under distribution shift, weighted and non-exchangeable variants give approximate bounds [57–59], and online methods adapt a threshold from feedback with long-run guarantees for arbitrary sequences [32, 33, 60, 61]. Conformal methods were applied to speech recognition [62], to the choice of how many hypotheses an LLM corrector of speech recognition output receives, with Learn-then-Test [29], and to LM outputs and abstention [63–65]. We apply CRC, LTT and online risk control to the commit decision of neural text entry, and we test them on three recording modalities.

Hallucination in LM-assisted decoding. LM stages can produce fluent text that the input does not support. In speech recognition this causes real harm [10], and generative error correction can replace correct words [66]. Reliability-oriented metrics that reward abstention on uncertain input are emerging for ASR [67].

Text-entry and BCI metrics. BCI work often reports information transfer rate [68] or CER and word error rate (WER). Utility metrics [69] and the practical bit rate [70] charge a speller for the time needed to correct its errors; text-entry research uses error metrics such as MSD and KSPC [71]. Studies with gaze-typing users quantify the time cost of selections and corrections [30], and LM-based prediction speeds up gaze typing [30, 72]. Our eWPM (Section 6) is close to the utility metric: it counts correct words per unit of total time, including corrections, and we report it next to an explicit no-BCI baseline.

Interoperability standards. Apple's BCI HID profile [20], W3C EMMA [21], Lab Streaming Layer [22], BIDS [23], Neurodata Without Borders [73], the BCI2000 and OpenViBE platforms [74, 75], IEEE P2731 terminology [76] and the IEEE P2794 reporting standard in development [77, 78] each cover part of the stack: OS input events, multimodal annotation, signal transport, data at rest, real-time experiment platforms, vocabulary and reporting. Benchmarks such as

MOABB and NeuralBench unify decoder evaluation [79, 80]. None of them defines a per-frame decoded-output format with confidence semantics.

3 Problem setting

Items, streams and candidates. An *item* is one attempt by the user to enter one sentence. We use "item" for the generic unit and keep the dataset's own word where it matters: a *trial* of the intracortical data, an *instance* (one participant typing one sentence) of the MEG data, and a composed *sentence* of the sEMG data. For item i, the decoder emits a stream of F_i frames: a matrix of log-probabilities $\log p_t(v)$ over a vocabulary of V tokens, $t = 1, \ldots, F_i$. A search over the stream with an LM returns a ranked list of K candidate sentences (the K-best list; $K = 10$ in all experiments) with scores, from which we compute $\hat{p}_{\text{top1}}$, the normalised probability of the top candidate.

Actions and losses. For each item, the system takes one action a_i: COMMIT the top candidate (the text is sent without further input); OFFER the top k candidates for the user to pick; OFFER_WORDS, an offer of the top candidate with uncertain words marked; or ABSTAIN, after which the user types the sentence with a fallback keyboard. Let $c_i \in \{0, 1\}$ indicate that the top candidate differs from the intended sentence (exact match after normalisation of case and punctuation), and let $f_i \in \{0, 1\}$ be the wrong-commit loss of the non-commit action ($f_i = 0$ for OFFER and ABSTAIN; for OFFER_WORDS, $f_i = 1$ if a wrong word is not marked). A *wrong commit* is a committed sentence that differs from the intended one.

The two error quantities. We report two rates over the n_{test} test items:

$$\text{wrong-commit rate} = \frac{\#\{\text{wrong commits}\}}{n_{\text{test}}} \approx \Pr(\text{commit} \wedge \text{wrong}), \tag{1}$$

$$\text{error among commits} = \frac{\#\{\text{wrong commits}\}}{\#\{\text{commits}\}} \approx \Pr(\text{wrong} \mid \text{commit}). \tag{2}$$

The first is the *marginal* risk: the rate at which wrong sentences reach the recipient per sentence attempted. It is the quantity that our risk controllers bound, because it is monotone in the commit threshold (Section 5.4). It can be made small by committing rarely, so we always report it together with the *commit rate*. The second is the *selective* risk that a user sees when looking at committed text. Our controllers do not bound it, and we report it next to every result.

Notation. Table 7 in Appendix A collects the symbols. $\alpha = 0.05$ is the target wrong-commit rate and $\delta = 0.1$ the failure probability of Learn-then-Test.

4 The NeuralTextStream format

NeuralTextStream (NTS) is the contract between a decoder and any consumer of its output (Figure 2). A stream of F frames over a vocabulary of V tokens holds an $F \times V$ array `log_probs` of natural-log probabilities (row t is a distribution at frame t), the frame times in seconds, the ordered vocabulary, its type (`char`, `key`, `word`, `phoneme`, `command` or `other`) and the index of the CTC blank (or `null` for non-CTC decoders). Optional fields give the frame period, the algorithmic latency, the producer and a `meta` object with named key prefixes (`device.*`, `decoder.*`, `lm.*`, `calibration.*`, `adapter.*`).

Table 1: Devices, data and streams. One adapter per device type maps decoder output into NeuralTextStream. All datasets are public. Test units are those of the trust evaluation.

	MEG (typing)	**Intracortical (speech)**	**Forearm sEMG (keys)**
Dataset (licence)	SpanishBCBL (CC BY-NC 4.0)	B2T '25 copy task (CC0)	Toronto sEMG (CC BY 4.0)
Participants	19 (healthy typists)	1 (T15, ALS)	19 (healthy typists)
Test unit	637 instances of 17 unique sentences	816 trials, 14 days	551 sentences composed from isolated presses
Decoder	own asynchronous ConvCTC	published online RNN; own phoneme-to-word decoder	own CNN
NTS vocabulary	`char` (CTC)	`phoneme` (CTC, 41 tokens)	`key` (W3C codes)
Frame	40 ms	80 ms	one per key press
Commit decision tested?	no (no right top-1)	yes	weakly (0–4% committed)

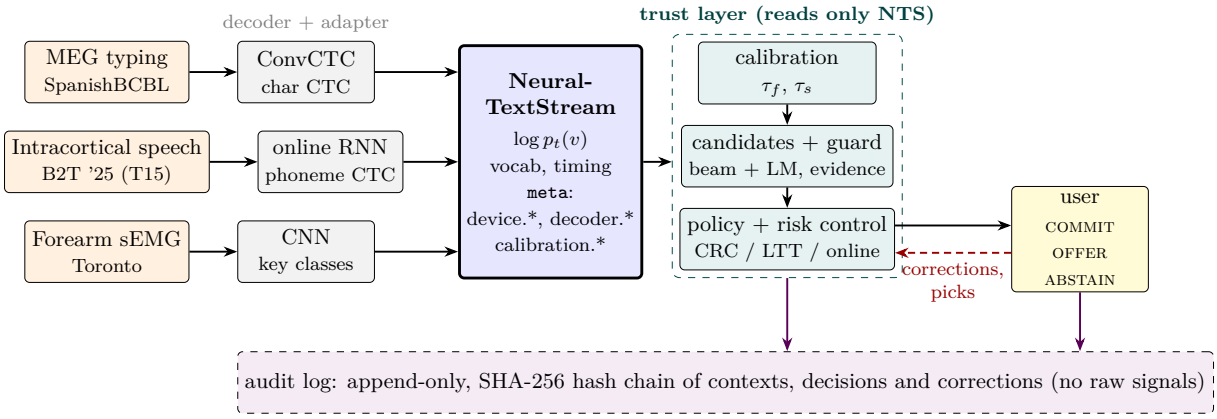


Figure 2: Architecture. Each device's decoder emits a NeuralTextStream through a small adapter. The trust layer reads only the stream: it calibrates confidence, builds candidate sentences with a neural-evidence guard, and chooses an action under a risk-control rule. The user's corrections and picks (dashed) feed the online re-calibration. Contexts, decisions and corrections can be written to an append-only, hash-chained audit log that stores no raw signals (Appendix E).

Unmodified probabilities. A producer must emit the decoder's own output distribution and must not sharpen or smooth it, for example for display. Calibration is left to the consumer, which can fit it on data with known truth. A field is interpreted as a probability only if the stream declares it as one in `calibration.*`. Ground-truth text must not appear in the stream (a leakage rule), and raw signals and direct identifiers stay out of it (privacy rules). Appendix E summarises the serialisations, the conformance suite and the audit log.

Three device types. Table 1 lists the adapters. Mapping the intracortical and sEMG decoders into NTS needed no change to the frame core of v0.1, a Development release written by the same authors. The trust harness needed one optional field (the truth in the stream's token space, because phoneme truth differs from the word truth that the user sees), and provenance added optional `meta` keys. The adapters have 148 (MEG), 412 (intracortical) and 568 (sEMG, including a file reader) lines of Python, with documentation. Every other part of the pipeline is the same code for all devices, but its parameters are re-fitted per device (Table 9).

5 The trust layer

The trust layer reads one NeuralTextStream per item and returns one action. It never reads raw signals or decoder internals. Figure 3 shows the decision flow. All parameters below are fitted per device on that device's fit data; Table 9 lists the fitted values.

5.1 Calibration

Decoder log-probabilities are often over-confident. We fit a frame temperature τ_f that rescales every frame, $\tilde{p}_t(v) \propto p_t(v)^{1/\tau_f}$, by minimising the cross-entropy of the true token at the greedy CTC peak frames of the fit streams. We then fit a sentence temperature τ_s on the scores of the K-best list, $\hat{p}_{\text{top1}} = \text{softmax}(\text{scores}/\tau_s)_1$, by minimising the binary negative log-likelihood of "the top candidate is right" [24]. Both are one-dimensional golden-section searches over $\log \tau \in [\log 0.05, \log 20]$. We report the expected calibration error (ECE, 15 equal-width bins) of $\hat{p}_{\text{top1}}$ before and after. The frame temperature is applied *before* the LM search, so it changes the decoded text as well as the confidence (Section 7.3).

5.2 Candidates and neural-evidence guard

Candidates come from a CTC prefix beam search with a character 6-gram LM trained on Tatoeba sentences, which returns a K-best list [40, 41]. For intracortical speech, a phoneme beam search is followed by a CMUdict lexicon, a Tatoeba word unigram and the same character LM (Section 6). Because the LM can write fluent text from any input, a guard scores how much of the top candidate the LM invented. It combines two signals: the share of the candidate's characters whose calibrated acoustic posterior is below 0.5 (*low evidence*), and one minus the posterior-weighted word agreement between the top candidate and the top five candidates (*K-best inconsistency*). The guard risk is the maximum of the two (both weights 1). A candidate whose guard risk exceeds a fitted limit cannot be committed. The guard is an optional component: it helped against signal-destroying inputs but not on real intracortical data (Sections 7.3 and 7.4).

Gate on a published sentence. For the published intracortical decoder we have the stream and the published LM-rescored sentence, but not the decoder's LM or weights. We put the published sentence on top of the K-best list of our own phoneme-to-word decoder, with a score equal to our best score plus a fixed bonus of 2.0, and give each published word the acoustic confidence of the aligned word in our top candidate (0 if no word aligns). The guard and the policy then run unchanged. This gate therefore needs the stream, the published text and a light reference decoder with its own lexicon and LM, but not the vendor's weights.

5.3 Decision policy

The policy chooses COMMIT, OFFER (with k candidates), OFFER_WORDS or ABSTAIN. Its thresholds (commit probability, maximum guard risk for a commit, offer size, abstain risk, fallback after an unused offer) are chosen on fit data from a grid to maximise eWPM (Section 6) subject to a wrong-commit rate of at most α on the fit data. This *plug-in* fit has no guarantee on new data, which motivates Section 5.4.

5.4 Finite-sample risk control for the commit threshold

The commit score s_i is the calibrated $\hat{p}_{\text{top1}}$ when the guard and the other policy rules allow a commit, and $-\infty$ otherwise. The layer commits when $s_i \geq \lambda$, and we use the loss

$$L_i(\lambda) = \max(c_i, f_i)\, \mathbf{1}[s_i \geq \lambda] + f_i\, \mathbf{1}[s_i < \lambda], \tag{3}$$

which upper-bounds the true loss and is non-increasing in λ, as conformal risk control requires. The controlled risk is $R(\lambda) = \mathbb{E}[L(\lambda)]$; with $f \equiv 0$, it is the probability that an item is committed *and* wrong (Section 3). On n calibration items with known truth, conformal risk control (CRC) [26] chooses

$$\hat{\lambda}_{\text{CRC}} = \inf\Big\{\lambda : \tfrac{n}{n+1}\hat{R}_n(\lambda) + \tfrac{1}{n+1} \leq \alpha\Big\}, \tag{4}$$

where $\hat{R}_n$ is the empirical risk. If the calibration items and a new item are exchangeable, then $\mathbb{E}[L_{n+1}(\hat{\lambda}_{\text{CRC}})] \leq \alpha$, where the expectation is over the draw of the calibration items and the new item. The implementation searches the finite calibration scores and $+\infty$ and commits on $s_i \geq \lambda$; this is valid and slightly conservative. CRC needs $L_i(+\infty) = f_i \leq \alpha$ on average; we use $f \equiv 0$ (no OFFER_WORDS under CRC), so "never commit" is always feasible. For $n < 1/\alpha - 1$ (19 items for $\alpha = 0.05$) no finite threshold is feasible and the layer never commits.

Learn-then-Test (LTT) [31] tests $H_0 \colon R(\lambda) > \alpha$ along a grid of thresholds from 1.00 down to 0.00 in steps of 0.01, fixed in advance, with exact binomial p-values and fixed-sequence testing, and stops at the first non-rejection. It gives $\Pr(R(\hat{\lambda}_{\text{LTT}}) \leq \alpha) \geq 1 - \delta$, where the probability is over the calibration draw and R is the population risk at the chosen threshold, not the rate on any finite test set.

Because L_i is non-increasing in λ, any threshold at or above $\hat{\lambda}_{\text{CRC}}$ has a loss no larger than that of $\hat{\lambda}_{\text{CRC}}$ on every item, so the bound still holds if a larger threshold is chosen on the same calibration data. We therefore deploy $\max(\lambda_{\text{tuned}}, \hat{\lambda}_{\text{CRC}})$, where λ_{tuned} is the plug-in threshold; the same holds for LTT. Where we report "pure" CRC, λ_{tuned} is not used. The guarantee also requires that the temperatures, the guard and the other policy thresholds are fitted on data disjoint from the calibration set. This holds in the random-split experiments (Section 7.5), where calibration items are drawn from the test half. It does *not* hold for the offline baselines of the drift experiment (Section 7.6), which were calibrated on fit data that also fitted the temperatures and the policy. **[TODO: R1: replace these baselines with the clean-calibration re-run and update this sentence.]** We use $\alpha = 0.05$ and $\delta = 0.1$ throughout.

5.5 Online re-calibration

Exchangeability fails when the error level changes during use. The online controller re-tunes the commit threshold from user feedback. Which feedback exists depends on the action: a corrected commit reveals a wrong commit; an uncorrected commit is right only if silence means right; an offer reveals the truth when the user picks or retypes; an abstention reveals nothing unless the final text is logged. The controller therefore accepts two feedback levels: the *realised loss* of every item (wrong commit or not), and, when the final text is known, the counterfactual commit loss of non-committed items. Feedback can arrive late. For commits, the realised loss requires that the user notices and corrects a wrong sentence; in our copy-task replays this comes from the known cue sentence. Error-related potentials are a possible implicit source [18]. We compare four methods (Table 12 in Appendix F gives the update rules and all assumptions). `window` runs CRC on the last 200 items with known truth, and `decay` runs weighted CRC on all past items with weights 0.99^{age}; neither has a guarantee under shift. `aci` updates the threshold as $\lambda_{t+1} = \max(0, \lambda_t + \eta(L_t - \alpha))$ with $\eta = 0.05$, and `aci_window` applies the same kind of update to the level of a windowed CRC. For any sequence of T items, both bound the average observed loss by α plus a term of order $1/T$, which is bounded in advance only if the scores are at most 1 and the non-commit loss is 0, and which grows with the feedback delay.

The two `aci` rules follow adaptive conformal inference and online risk control [32, 33, 61]. All online settings were fixed before the first evaluation run. The bounds are long-run averages over the T items of a sequence; they say nothing about short windows. A wrong commit that the user does not notice (a *silent error*) is outside them.

6 Evaluation protocol

6.1 Effective words per minute

Error rates compare decoders but do not tell whether a user saves time. We therefore simulate a user who reads the system output, accepts or corrects a commit, picks from an offer, or types the sentence with a fallback keyboard. For item i with intended sentence y_i, the simulation

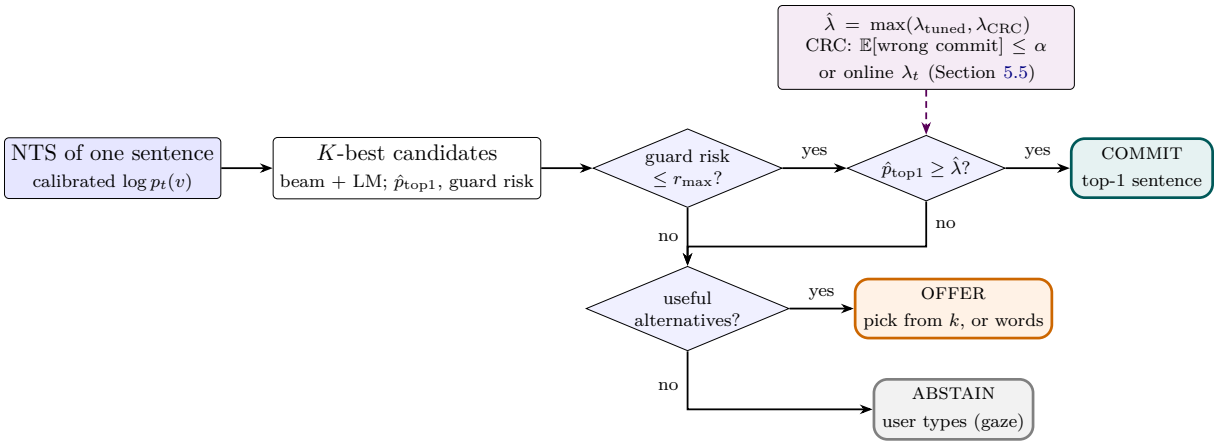


Figure 3: Trust decision for one item. Candidates come from the calibrated stream. A candidate is committed only if the neural-evidence guard accepts it and its calibrated confidence reaches the threshold $\hat{\lambda}$, which is the larger of the eWPM-tuned threshold and the CRC threshold (or the online threshold λ_t). Otherwise the layer offers alternatives or abstains, and the user types with the fallback keyboard.

returns the total time t_i (input, reading, selections, corrections and retyping) and the final text. The effective words per minute are

$$\text{eWPM} = 60 \cdot \frac{\sum_i \mathbf{1}[\text{final text}_i = y_i] \ \text{words}(y_i)}{\sum_i t_i}, \tag{5}$$

a ratio of sums over all test items in which a sentence counts only if it is exactly right at the end. Like the BCI utility metric [69], eWPM charges the time needed to fix errors. A *silent error* is a wrong final text that the user did not correct; it reaches the recipient and counts zero words. eWPM does not price the harm of a silent error beyond the lost words, so we report the silent-error rate separately.

6.2 User model and baselines

The user model is a gaze-typing user. The fallback keyboard runs at 20 characters per minute, and scanning and selecting one of $k = 3$ word alternatives takes 4.5 s; both are matched to measurements of one eye-gaze typing user with ALS [30]. All other time costs are assumptions (Table 8). The neural input runs at a device-specific speed: the measured 159.3 characters per minute of the intracortical participant; 75.8 characters per minute for sEMG (the 0.792 s median press period); and 152 characters per minute for MEG (the healthy participants' typing speed).

Silent errors and the asymmetry between commit and offer. After a COMMIT, the user reads the sentence and misses each wrong word independently with probability 0.05; a missed wrong word becomes a silent error. After an OFFER, the user compares the candidates on purpose and sees every error, and chooses the cheapest of picking, fixing words or retyping, using the true costs. This asymmetry favours offers. It encodes the hypothesis that committed text is checked less carefully than offered text, which is not measured. The model does not make fluent wrong sentences harder to notice than garbled ones: a sentence with more wrong words is more likely to keep at least one of them. Section 7.2 reports how the MEG conclusions change under other user-model settings. **[TODO: R1: user-model sensitivity grid on all three devices, a symmetric-offer variant and a misspecified-user evaluation without refitting.]**

Population mismatch. The fallback and selection costs are those of an eye-gaze user with ALS. This matches the intracortical participant, who has ALS. The MEG and sEMG decoders,

however, read overt typing by able-bodied people, who would not need a gaze keyboard. For these two devices, eWPM answers a hypothetical question: how fast would a gaze-typing user be with a decoder that has this error profile at this input rate?

Baselines. *Gaze typing only*: the same user types every sentence with the fallback keyboard and no BCI. This baseline is error-free and has no reading or correction time, which slightly favours it, but it also has no LM-based word prediction, which favours the BCI arms; LM prediction is known to speed up gaze typing [30, 72]. **[TODO: R1: LM-assisted gaze baseline (same 6-gram, word completion with $k = 3$ slots).]** *Always commit*: commit the top candidate of every item. *Always offer*: offer the top candidates of every item and never commit; this is what most BCI spellers and AAC systems do.

6.3 Datasets and splits

Intracortical speech (Brain-to-Text '25). The public copy-task file holds the online phoneme CTC logits, the cue sentences and the published LM-rescored sentences of participant T15, an implant recipient in the BrainGate2 clinical trial, for 1,718 trials over 15 days [2, 39]. We use the 1,628 trials with the 125,000-word vocabulary, which span 14 days; the 90 trials with a 50-word vocabulary are excluded. The file has no trial time stamps; we take the file order within each day as the trial order. We fit on the first half of each day (812 trials) and test on the second half (816 trials); one sentence occurs in both halves. Inside the fit half, the frame and sentence temperatures use the first half of the fit trials and the policy uses all of them. We evaluate two decoders: our own phoneme-to-word decoder (phoneme beam, CMUdict lexicon, Tatoeba word unigram and character LM; test WER 0.149) and the published sentence (test WER 0.043), which we gate as described in Section 5.

Forearm sEMG (Toronto). The dataset holds isolated key presses from 19 participants over two days [8, 81]. Because it has no continuous typing, we compose sentences from single presses, with oracle press timing and no inter-key context: one real press window per character, at the median press period. Sentences are held-out Tatoeba sentences that the LM did not see; exact overlap is excluded, but near-duplicates were not checked, and the style matches the LM corpus. Each participant contributes 30 fit sentences built from presses of the first day-2 recording of each letter, and up to 30 test sentences from later recordings (570 fit, 551 test). With about 10 presses per letter per pool, a press can appear in several test sentences, so sentence-level intervals are too narrow and participant intervals are primary. A CNN classifies each press. Three conditions vary how much of the user's data the decoder saw: *session-calibrated* (day 1 plus half of the day-2 recordings, cross-fitted: the decoder that predicts the test presses is trained on the first recordings, and the fit sentences are predicted by the decoder trained on the later ones), *cross-session* (day 1 only) and *cross-subject* (4-fold leave-participants-out). Within the fit sentences, one half (A) fits the temperatures and the LM weight and the other half (B) fits the policy; CRC and LTT use the last 142 sentences of half B and the grid the first half.

MEG typing (SpanishBCBL). Healthy participants typed memorised Spanish sentences in an MEG scanner [5, 82]. We use 19 participants and split by a hash of the sentence text into 101 training, 11 validation and 17 test sentences. The split is within participants: the same 19 people appear in training and test, with different sentences. No test sentence appears in training or in the LM text. The test set has 637 instances of the 17 sentences; all trust parameters are fitted on the 437 validation instances of 11 sentences. Our asynchronous ConvCTC decoder (convolutional front end and 4 Conformer layers of width 256 [83]) receives the MEG segment from 0.4 s before the first key press to 0.5 s after the last, its length and the participant index, but no key times at inference. It uses key times as auxiliary training labels, and the sentence

bounds come from the keyboard log, so it is not a free-running decoder and cannot be trained for a user who cannot type (Appendix D). The trust harness decodes its streams with a fixed beam setting (LM weight 0.5, frame temperature fitted on validation) whose top-1 CER is 0.522, higher than the 0.475 of the beam setting tuned on validation CER (Table 2). A synchronous decoder with oracle key times gives the same trust results (Appendix D).

6.4 Controls

Decoder controls show that the decoders use the neural signal: shuffled MEG, zero input, and a decoder trained with permuted labels. *Trust-layer controls* come in two classes. *Signal-destroying controls* keep the stream format but remove the neural information: frame-shuffled streams, near-uniform streams, zero-input decoder output and a permuted-label decoder. A stream-only layer should not commit these. The *intent-mismatch control* is the real stream of a different sentence by the same participant. It carries genuine neural evidence, only for a sentence the user did not intend now, so a stream-only layer cannot detect it.

6.5 Statistics

We report 95% percentile bootstrap confidence intervals and paired differences on the same items. The resampling unit is stated with every interval: trials `[trial]` and days `{day}` for Brain-to-Text '25; sentences `[sent]` and participants `{participant}` for sEMG, where participants are primary; for MEG trust results, unique sentences (cluster bootstrap over the 17 test sentences), participants, and both at once (two-way), with the two-way interval primary for comparisons with gaze typing. MEG decoder CER intervals over instances are too narrow and are not used for inference. We name two primary endpoints per device: the wrong-commit rate of the trust layer and the paired eWPM difference between the trust layer and gaze typing. All other intervals are descriptive and not corrected for multiplicity. Decoder settings, user-model values, the policy grid, α, δ and the online step sizes were fixed in the experiment configurations before the evaluation runs. The "noticed errors only" online runs were added after the first run without changing any setting. Every decoder and trust run uses one seed and one fit/test split. **[TODO: R1: decoder seeds and repeated fit/test splits; sentence-cluster CIs for MEG CER.]**

7 Results

7.1 The decoders use the neural signal

Before we evaluate the trust layer, we check that each decoder reads its input and how often its top sentence is right (Table 2). Our asynchronous MEG decoder reached a test CER of 0.506 with greedy decoding and 0.475 with a beam search and character LM tuned on validation CER. With shuffled MEG the greedy CER rose to 0.760, and with zero input to 0.925. A text-only output that ignores the MEG (one fixed training sentence) reaches a CER of 0.688 on the same test instances, so the MEG accounts for the step from about 0.69 to 0.51, not from 1.0. Word error rates stayed high (0.842 greedy, 0.809 beam), and in the trust harness the top-1 sentence was exactly right for 2 of 637 instances. Six of the 17 test sentences have a close variant in the training set; excluding them gave 0.519 (greedy) and 0.492 (beam), so memorisation of near variants explains at most 0.02 of the CER. We selected the training schedule on validation CER; a longer schedule gave the same validation CER and is not reported further. For context, the published asynchronous encoder reports 0.59 ± 0.02 on SpanishBCBL with another split and without key-time training labels [6], and the published synchronous decoder 0.29 [5]. Our MEG results describe one weak decoder on 17 unique test sentences, not MEG decoding in general.

On Brain-to-Text '25, our phoneme-to-word decoder reached WER 0.149 with the right sentence first for 39% of test trials, and the published decoder WER 0.043 with 81%. On

Table 2: Decoder quality and input controls (test sets). Lower CER/WER is better. Exact: share of items whose top-1 sentence is exactly right; in K-best: share whose intended sentence is in the 10-best list. MEG: the beam row tuned on validation CER is the decoder result; the trust harness uses a fixed beam setting (Section 6). "—": not applicable or not computed.

Device / decoder	CER	WER	Exact / in K-best	Controls and floors
MEG, asynchronous, greedy	0.506	0.842	—	shuffled 0.760, zeros 0.925
MEG, beam + LM tuned on validation	0.475	0.809	—	text-only floor 0.688
MEG, beam in the trust harness	0.522	—	0.003 / 0.016	—
MEG, 11 sentences without close variant	0.519 / 0.492	—	—	(greedy / beam)
B2T '25, our phoneme-to-word decoder	0.095	0.149	0.390 / 0.623	—
B2T '25, published decoder	0.031	0.043	0.806 / —	—
sEMG, key accuracy (balanced) session-cal. / cross-session / cross-subject	0.299 (0.367) / 0.168 (0.240) / 0.209 (0.226)		—	permuted 0.022 (0.032); majority 0.338, uniform 0.037
sEMG, sentences after LM	0.123 / 0.398 / 0.479	0.214 / 0.579 / 0.681	0.459 / 0.118 / 0.038	in K-best 0.64 / 0.21 / 0.08

sEMG, the CNN reached key accuracies of 0.299 (session-calibrated), 0.168 (cross-session) and 0.209 (cross-subject). These are below the majority-class rate (space, 0.338), but balanced accuracies (0.367, 0.240, 0.226) are well above uniform chance (0.037); with permuted training labels, accuracy fell to 0.022 and balanced accuracy to 0.032. After the LM, the right sentence was first for 46%, 12% and 4% of composed sentences.

7.2 The commit decision: trust layer vs always commit and always offer

Table 3 and Figure 1 give the main result. The comparison that matters for a speller is with always offering: the trust layer can only gain speed by committing, and it pays for each commit with some risk of a wrong sentence.

Intracortical speech, our decoder. Always committing the top sentence gave 11.45 eWPM and a wrong-commit rate of 0.610. The trust layer, on the same output, gave 11.55 eWPM and a wrong-commit rate of 0.059 $[0.043, 0.075]$. It committed 23% of trials, and 48 of these 189 commits were wrong (error among commits 0.25). The paired eWPM difference to always commit was $[-0.11, +0.30]$ over trials and $[-0.17, +0.33]$ over days, so the data exclude a slowdown larger than about 1.5%. Always offering gave 11.37 eWPM with no wrong commits. The commit decision therefore bought +0.18 eWPM (1.6%) over always offering, at a cost of 59 wrong commits per 1,000 sentences, of which 2 per 1,000 were silent. **[TODO: R1: paired CI of trust minus always offer (always-offer analysis).]**

Intracortical speech, gate on the published sentence. Always committing the published sentence gave 19.61 eWPM and a wrong-commit rate of 0.194, although its WER is only 0.043. The gate reduced the wrong-commit rate to 0.044 $[0.031, 0.059]$, at a speed cost of 4.3% (paired eWPM $[-0.98, -0.72]$). It committed 47% of trials, and 36 of 386 commits were wrong (0.09). Always offering gave 18.11 eWPM, so the gate was 0.66 eWPM (3.6%) faster than always offering. All policies on this device were far faster than gaze typing alone (4.08 eWPM; trust minus gaze $[+6.96, +7.97]$, gate minus gaze $[+13.81, +15.53]$ over trials).

Forearm sEMG (composed sentences). In all three decoder conditions, the trust layer reduced the wrong-commit rate from 0.541, 0.882 and 0.962 to 0.004, 0.004 and 0 (2 of 551 sentences in the first two conditions). It committed 21, 10 and 0 of 551 sentences, with 2, 2 and 0 of them wrong. The policy therefore acted almost as an always-offer policy, and its speed gain over always commit (participant CIs $\{+0.01, +0.41\}$, $\{+0.46, +0.69\}$ and $\{+0.44, +0.67\}$) came from offers, from which the user picks the right sentence when it is in the list. Only the session-calibrated decoder clearly beat gaze typing (8.02 vs 4.27 eWPM; trust minus gaze $\{+2.66, +4.85\}$ over participants). With the cross-session decoder, the trust layer was 0.41 eWPM above

Table 3: Safety and speed on three device types (test sets). eWPM with 95% bootstrap CIs; square brackets: trials (B2T '25), sentences (sEMG) or unique-sentence clusters (MEG); curly brackets: days (B2T '25) or participants (sEMG, MEG; primary for sEMG). Wrong-commit: wrong commits / all items. Commit: commits / all items. Err. | commit: wrong commits / commits (counts in brackets). Silent: silent errors / all items. Always offer: never commit (B2T '25 point values from the per-trial item tables; on MEG the trust policy offered on every instance and is the always-offer policy). "—": no commits.

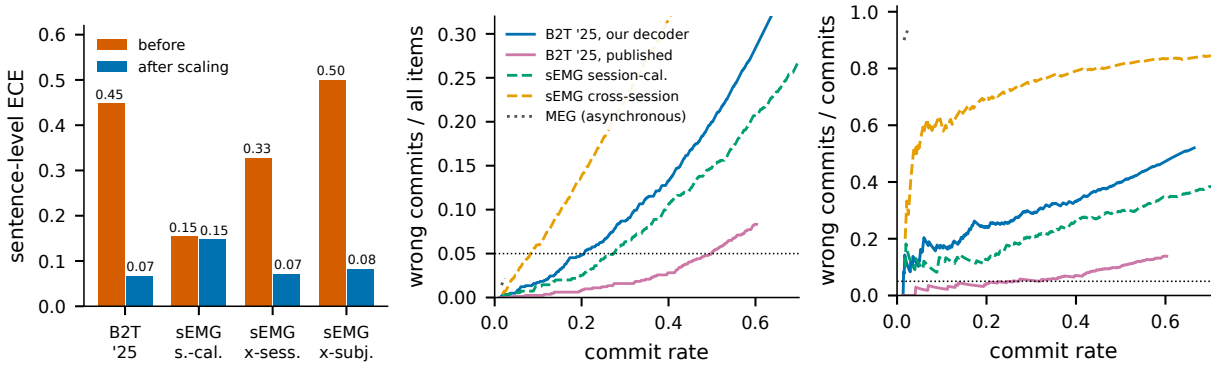
Device / condition	Policy	eWPM [CI] {CI}	Wrong-commit	Commit	Err. | commit	Silent
B2T '25, our decoder	always commit	11.45 [10.95, 12.04] {10.72, 12.32}	0.610	1.00	0.61	0.044
	trust	11.55 [11.03, 12.06] {10.85, 12.38}	0.059	0.23	0.25 (48/189)	0.002
	always offer	11.37	0	0	—	0
B2T '25, published	always commit	19.61 [18.68, 20.51] {17.76, 21.45}	0.194	1.00	0.19	0.025
	trust gate	18.77 [17.88, 19.62] {17.05, 20.46}	0.044	0.47	0.09 (36/386)	0.004
	always offer	18.11	0	0	—	0
B2T '25	gaze typing only	4.08 [4.04, 4.12]	0	—	—	0
sEMG session-cal.	always commit	7.81 [7.36, 8.30] {6.60, 9.01}	0.541	1.00	0.54	0.060
	trust	8.02 [7.63, 8.45] {6.90, 9.15}	0.004	0.04	0.10 (2/21)	0
sEMG cross-session	always commit	4.10 [3.89, 4.34] {3.59, 4.70}	0.882	1.00	0.88	0.149
	trust	4.68 [4.52, 4.86] {4.21, 5.25}	0.004	0.02	0.20 (2/10)	0
sEMG cross-subject	always commit	3.53 [3.37, 3.72] {3.19, 3.88}	0.962	1.00	0.96	0.167
	trust	4.08 [3.98, 4.20] {3.83, 4.36}	0	0	—	0
sEMG	gaze typing only	4.27	0	—	—	0
MEG (asynchronous)	always commit	2.580 [2.41, 2.76] {2.41, 2.79}	0.997	1.00	1.00	0.235
	trust (= always offer)	3.203 [3.07, 3.36] {3.08, 3.37}	0	0	—	0
MEG	gaze typing only	3.155 [3.02, 3.29]	0	—	—	0

gaze typing over sentences $[+0.26, +0.56]$, but not over participants $\{-0.04, +0.96\}$. With the cross-subject decoder it was below gaze typing (4.08 vs 4.27; [-0.28, -0.09] over sentences, $\{-0.43, +0.08\}$ over participants). **[TODO: R1: always-offer rows for sEMG.]**

MEG (negative control for the commit decision). The top sentence was right for 2 of 637 test instances, and the intended sentence was in the 10-best list for 1.6%. Always committing gave a wrong-commit rate of 0.997 and 2.580 eWPM. The trust layer committed nothing and offered candidates on every instance, so it was the always-offer policy: 3.203 eWPM. Gaze typing alone gave 3.155. The paired difference to gaze typing was +0.05 eWPM, with 95% CIs of [-0.04, +0.15] over sentences, [-0.07, +0.22] over participants and [-0.11, +0.26] two-way. The two-way interval lies within ±0.32 eWPM (10% of gaze speed), but we set this margin after seeing the data, so we read the result as "no detectable difference", not as equivalence. The trust layer was above gaze typing for 10 of 19 participants. Its advantage over always commit (+0.62 eWPM, [+0.49, +0.75]) depends on the unmeasured miss probability: with a miss probability of 0, always commit gave 3.24 and the trust layer 3.20 (Table 11). Two other plausible settings reverse the comparison with gaze typing: with a 40 characters-per-minute fallback (the measured speed of a second gaze user in [30]), gaze typing gave 6.31 and the trust layer 5.15 eWPM; with selection, scanning and navigation times doubled, 3.15 and 3.00 (trust minus gaze [-0.23, -0.08] over sentences).

7.3 Confidence predicts errors; calibration fixes sentence-level ECE but can hurt decoding

On Brain-to-Text '25, the calibrated top-1 probability predicted a wrong top sentence with AUROC 0.797 at the word level and 0.884 at the phoneme level; per word, the acoustic word confidence gave AUROC 0.844. On sEMG, AUROC was 0.83, 0.76 and 0.67 in the three conditions. We did not compute CIs for these AUROCs. The raw intracortical logits were over-confident (frame temperature 3.78), and sentence ECE fell from 0.448 to 0.067. The same temperature made the ECE of the per-frame phoneme peaks worse (0.090 to 0.130): the temperature that is optimal for the likelihood is not optimal for the peak confidences. On sEMG, sentence ECE fell



(a) Sentence-level ECE. **(b)** Marginal risk. **(c)** Selective risk.

Figure 4: Calibration and the commit trade-off. (a) Expected calibration error of the sentence confidence before and after temperature scaling (test sets). (b) For each commit threshold on the calibrated score: share of all test items committed (x) and committed wrongly (y). (c) The same thresholds: error among commits (wrong commits / commits). Items blocked by the guard or other policy rules are never committed. Curves start at 10 commits. Dotted line: $\alpha = 0.05$. The cross-subject sEMG condition is absent because no candidate passed the guard; on MEG all but a few commits are wrong.

from 0.154 to 0.148, from 0.328 to 0.072 and from 0.500 to 0.083 (frame temperatures 1.25, 1.78 and 2.17; Figure 4a). **[TODO: R1: reliability diagrams before and after calibration (harness re-run with per-bin output).]**

Because the frame temperature is applied before the LM search, it also changes the decoded text. On sEMG, the LM weight was tuned on raw streams; the flatter calibrated streams give the LM relatively more weight, and the CER of the top sentence rose from 0.369 to 0.479 (cross-subject), 0.387 to 0.398 (cross-session) and 0.116 to 0.123 (session-calibrated). Without calibration, the cross-subject trust layer reached 4.76 eWPM [4.60, 4.94] {4.25, 5.35} with a wrong-commit rate of 0.005, above gaze typing (4.27); with calibration it reached 4.08, below gaze typing. Without calibration, the session-calibrated layer was faster (8.15 vs 8.02 eWPM) but had a wrong-commit rate of 0.045 instead of 0.004. On Brain-to-Text '25, removing calibration raised the wrong-commit rate from 0.059 to 0.071 at the same speed (11.57 vs 11.55 eWPM).

The guard did not help on real intracortical data: without it, the wrong-commit rate fell from 0.059 to 0.036 at the same speed (11.51 eWPM), and its AUROC for a wrong top sentence (0.702) was below that of $\hat{p}_{\text{top1}}$ (0.797). On sEMG its AUROC was 0.65, 0.56 and 0.50. The guard's value lies in the signal-destroying controls (Section 7.4). **[TODO: R1: unified ablation over all devices and conditions (calibration applied to the beam vs to the confidence only, guard, CRC), and LM weight re-tuned after the frame temperature.]**

Figures 4b and 4c show the trade-off as the commit threshold varies. For the marginal risk (Figure 4b), the published decoder can commit about half of its sentences at the 0.05 level, and our weaker decoder about a fifth. For the selective risk (Figure 4c), the error among commits of our decoder falls to about 0.1–0.2 only when fewer than about 10% of trials are committed and rises to about 0.5 at a commit rate of 0.6, while that of the published decoder stays near or below 0.15. On sEMG cross-session, most commits are wrong at all but the lowest commit rates. A rule that controls the marginal risk can therefore accept a high error among commits on a weak decoder.

7.4 Controls: signal-destroying streams are not committed; intent mismatch is not detectable

Always commit wrote a sentence for every control stream, and on MEG 85–100% of these sentences were fluent Spanish (85% on real input). The trust layer committed none of the signal-destroying controls on Brain-to-Text '25 (frame-shuffled and uniform streams, 300 trials) or on

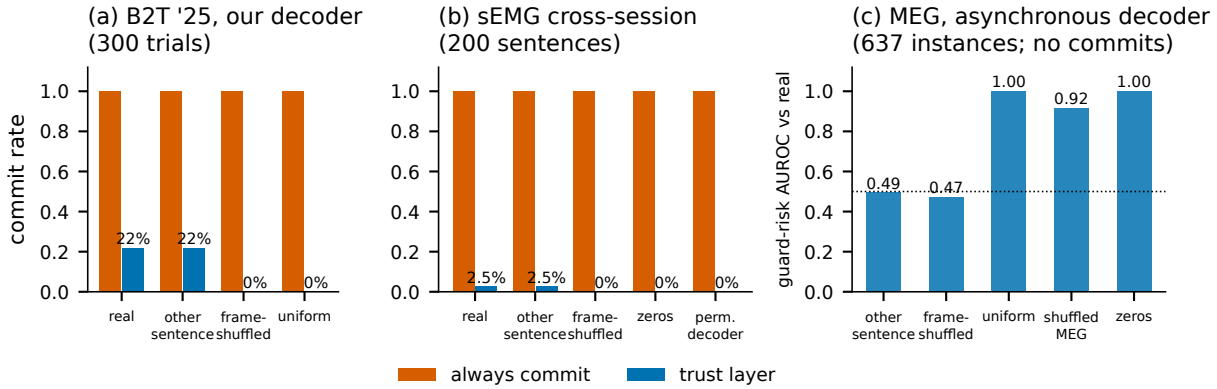


Figure 5: Controls. (a, b) Commit rate of always commit (orange) and the trust layer (blue) on real input, the intent-mismatch control (other sentence) and signal-destroying controls. (a) B2T '25, our decoder, 300 random test trials. (b) sEMG cross-session, 200 random test sentences. For "real" and "other sentence", commits are a mix of right and wrong; for all other inputs any commit is wrong. (c) MEG, asynchronous decoder, all 637 test instances: AUROC of the guard risk for separating each control from real input (0.5: no separation). The MEG trust layer committed nothing on any input, including real input.

sEMG (frame-shuffled, zero-input and permuted-decoder streams, 200 sentences per condition) (Figure 5a, b). This is an informative test on these two devices, because the same layer committed 22% (Brain-to-Text '25) and 2.5–4% (sEMG) of the real inputs in the same audit. On Brain-to-Text '25, the guard risk separated frame-shuffled and uniform streams from real ones with AUROC 0.981 and 0.998. On MEG the trust layer committed nothing on real input either, so its zero commits on control streams carry no information. The guard risk is the informative MEG result (Figure 5c): it separated shuffled MEG, zero input and uniform streams from real streams (AUROC 0.92, 1.00 and 1.00), but not frame-shuffled streams (0.47), whose per-frame evidence for this weak decoder looks like that of real streams.

The intent-mismatch control behaves differently. Given the real stream of another sentence, the trust layer on our Brain-to-Text '25 decoder committed 22% of items, the same rate as on matched input, and on sEMG it also committed at the matched rate. This is expected: the stream is genuine neural evidence, only for a sentence the user did not intend, and nothing in the stream reveals the intent. The gate on the published sentence committed none of these items, but only because its text came from the matched trial and disagreed with the mismatched stream; this is a property of the test construction, not a capability of the gate.

7.5 Conformal risk control bounds the mean marginal risk, not the error among commits

The plug-in policy exceeded the target on our Brain-to-Text '25 decoder (0.059 on the full test set). We drew 200 random calibration/test splits of the 816 test trials, with 408 trials for calibration (Table 4). A random split of one fixed pool is exchangeable by construction, so the CRC theorem guarantees a mean over splits of at most α; the observed means of 0.0483 (our decoder) and 0.0478 (published gate) confirm the implementation rather than test the assumption. They came at eWPM changes of -0.3% and $+0.06\%$ relative to the fixed plug-in gate. Individual splits vary: the test rate exceeded 0.05 in 41–45% of CRC splits, and part of this spread is test-sampling noise (standard error about 0.011 with 408 test items). LTT gave a mean of 0.0345 for both decoders and a 90th percentile of the test rate of 0.0515 (ours) and 0.0542 (published), against 0.0686 for CRC. Because the 816 trials are the population of this experiment, we can compute the calibration-conditional risk that LTT bounds: the wrong-commit rate of the chosen threshold on all 816 trials. It exceeded α in 0.5% (ours) and 1.5%

Table 4: CRC and LTT for the commit threshold ($\alpha = 0.05$, $\delta = 0.1$; 408 calibration items; 200 random splits of the test pool). Wrong-commit: mean test rate over splits; p90: its 90th percentile; $P_{>\alpha}$: share of splits with a test rate above α; $\text{Pool}_{>\alpha}$: share of splits in which the chosen threshold's rate on the whole pool exceeds α (the calibration-conditional risk that LTT bounds with probability $1 - \delta$); Err. | commit: mean wrong-commit / mean commit rate. Fixed gate: the plug-in policy fitted on the fit data. MEG: asynchronous decoder, pool of 637 instances.

Data (n_{test})	Method	Wrong-commit	p90	$P_{>\alpha}$	$\text{Pool}_{>\alpha}$	Commit	Err. | commit	eWPM
B2T '25, our decoder (408)	fixed gate	0.059	0.069	0.86	1.00	0.233	0.25	11.57
	CRC	0.048	0.069	0.45	0.33	0.194	0.25	11.53
	LTT	0.035	0.052	0.14	0.005	0.156	0.22	11.52
B2T '25, published gate (408)	fixed gate	0.044	0.054	0.20	0	0.471	0.09	18.77
	CRC	0.048	0.069	0.41	0.29	0.479	0.10	18.78
	LTT	0.035	0.054	0.14	0.015	0.427	0.08	18.70
MEG, asynchronous (229)	fixed gate	0	0	0	0	0	—	3.204
	CRC	0.019	0.031	0	0	0.021	0.92	3.199
	LTT	0.021	0.031	0	0	0.023	0.93	3.199

(published) of LTT splits, within $\delta = 0.1$, and in 33% and 29% of CRC splits, which CRC does not promise to avoid.

Neither method bounds the error among commits. Under CRC, 25% of commits of our decoder were wrong (10% for the published gate), and under LTT 22% (8%). On MEG, where almost every possible commit is wrong, CRC still committed 2.1% of instances, because this is within the budget, and 92% of these commits were wrong (LTT: 2.3% committed, 93% wrong). LTT committed slightly more than CRC on MEG because its fixed grid extends to $\lambda = 0$, while CRC only considers thresholds at calibration scores, so test items scoring below the lowest calibration score commit under LTT but not under CRC. Neither gained speed (eWPM 3.199 vs 3.204 for the plug-in gate, which committed nothing). Calibration splits grouped by sentence (11 of 17) or by participant (12 of 19) gave the same means (CRC 0.0194 and 0.0201). This is control without utility: the marginal bound is met by spending the budget on commits that are almost always wrong. The deployable rule $\max(\lambda_{\text{tuned}}, \hat{\lambda}_{\text{CRC}})$ inherits the plug-in threshold, which committed nothing on MEG. On sEMG, with 142 calibration sentences, both methods were feasible (cross-session CRC threshold 0.204, bound 0.049), but the plug-in thresholds were stricter, so the combined rule equalled the plug-in policy. With 100 calibration trials, CRC on our decoder still had a mean of 0.0464, but the 95th percentile of the test rate rose to 0.094, and LTT committed only 9.7% of trials. **[TODO: R1: calibration sets of 20–50 trials and calibration within a day (small-n CRC analysis).]**

7.6 Online re-calibration under a shift between the halves of each day

When we calibrated CRC on the first half of each Brain-to-Text '25 day and applied the threshold to the second half in file order, the test wrong-commit rate was 0.0625 (Table 5). On sEMG cross-session, pure CRC calibrated on fit data gave 0.0635. Two caveats apply. First, both calibration sets had also fitted the temperatures and the policy grid (Section 5.4), so part of the excess may be a reuse effect. **[TODO: R1: offline CRC with calibration data disjoint from the data that fitted temperatures, guard and policy (clean drift baseline); rename the phenomenon if the excess disappears.]** Second, when we shuffled the order of the test half, offline CRC still gave 0.0625, so the excess is a level difference between the calibration data and the test half, not a trend within the test half. We therefore call it a shift between halves, not within-session drift.

The four online methods brought the rate to 0.040–0.053 on both data sets, at an eWPM cost of at most 0.6%. `aci_window` ended below 0.05 in both cases (0.0478 and 0.0490). `aci` ended at 0.0502 and 0.0526: the update drives the long-run rate towards α, not below it. Its certificate after the fact equalled these values and was within the a-priori bounds of 0.064 (Brain-to-Text

Table 5: Wrong-commit rate when calibration data and test data differ in level. Offline CRC: calibrated on the first half of each day (B2T '25) or on fit data (sEMG cross-session, pure CRC), then fixed. Online methods: warm start on the same data, then the test data in file order. Shuffled: test order randomised (mean of 20). Certificate / a priori: the `aci` bound after the fact and in advance (Table 12). Single runs; trial order is the file order (the data have no time stamps). **[TODO: R1: clean-calibration offline baseline; CIs over days.]**

Data	Offline CRC	`window`	`decay`	`aci`	`aci_window`	`aci` cert. / a priori
B2T '25, ours (816)	0.0625	0.0453	0.0417	0.0502	0.0478	0.0502 / 0.0637
B2T '25, ours, shuffled	0.0625	0.0502	0.0444	0.0510	0.0492	—
B2T '25, published (816)	0.0490	0.0429	0.0392	0.0478	0.0453	0.0478 / 0.0593
sEMG cross-session (551)	0.0635	0.0508	0.0399	0.0526	0.0490	0.0526 / 0.0809
eWPM, B2T '25, ours	11.553	11.490	11.500	11.510	11.489	

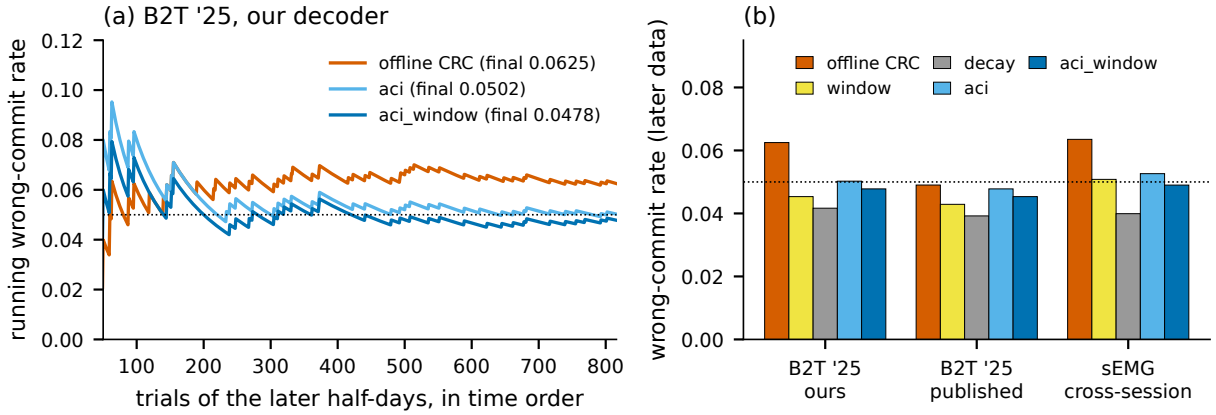


Figure 6: Online re-calibration keeps the long-run rate near the target when calibration and test data differ in level. (a) Running wrong-commit rate over the second half-days of B2T '25 (our decoder, 816 trials in file order; the data have no time stamps), recomputed from the committed per-trial items; final values equal Table 5. The first 50 trials are not shown. (b) Final wrong-commit rate per method and data set; single runs, no uncertainty shown. Dotted line: $\alpha = 0.05$. The offline baselines were calibrated on data that also fitted the temperatures and the policy.

'25, 816 trials) and 0.081 (sEMG, 551 sentences). These a-priori bounds are weaker than the 0.0625 of offline CRC, so at this sequence length the guarantee is loose. The `window` and `decay` methods gave 0.0399–0.0508 without any guarantee. With the test order shuffled, the online methods gave 0.0444–0.0510. Feedback delayed by 10 trials gave 0.0404–0.0502 on Brain-to-Text '25. Without a calibration block (cold start over all 1,628 trials), `aci` gave 0.0442 and `aci_window` 0.0491, but `window` gave 0.0516. On the published gate, offline CRC was already below the target (0.0490) and the online methods gave 0.0392–0.0478.

No method controlled short windows. The maximum wrong-commit rate over 50-trial windows was 0.10–0.12 on our Brain-to-Text '25 decoder and up to 0.18 on the published gate, against a median of 0.120 and a 95th percentile of 0.180 for an i.i.d. Bernoulli(0.05) sequence of the same length. When only noticed errors were fed back, `aci` gave 0.0515 on Brain-to-Text '25 and 0.0635 on sEMG cross-session, while its certificate on the observed losses stayed at 0.0496 and 0.0518: silent errors look correct to the controller. On MEG, in participant order (a sequence across participants, not a session), the online methods gave 0.0078–0.0204, and every commit was wrong.

8 Discussion

What the commit decision is worth. Against always committing, the trust layer looks like a large safety gain at no speed cost. Against always offering, which is what most spellers do, the picture is more modest. On our intracortical decoder, committing bought 1.6% speed for 59 wrong commits per 1,000 sentences; on the published decoder, 3.6% for 44 per 1,000. Whether this trade is worth it depends on the user and the context, and it should be a setting, not a default. The layer's main value on these data is that it makes the trade explicit and bounded: the user or clinician chooses α, and the layer reports the commit rate and the error among commits that follow from it. On sEMG and MEG, the layer almost never committed, so it reduced to an always-offer policy; on MEG, this was the correct behaviour for a decoder whose top sentence was almost never right. The layer cannot create information that the decoder does not have; it can only stop the LM from presenting a guess as an answer.

Marginal vs selective risk. The controlled quantity, the probability that an attempted sentence is committed and wrong, bounds how often wrong text reaches the recipient per sentence attempted. It does not bound how often committed text is wrong: one commit in four on our intracortical decoder, and nine in ten under CRC on MEG. A user who sees committed text may care more about the second quantity. We chose the marginal risk because it is monotone in the threshold, which CRC needs, and because it is the right budget for harm per attempt. A controller for the selective risk, for example LTT with binomial tests conditional on the number of commits and a minimum commit rate, is a natural complement. **[TODO: R1: selective-risk controller results (LTT on error among commits) on B2T '25.]**

Strong decoders still need a gate. The published Brain-to-Text '25 decoder has a WER of 0.043, yet it committed a wrong sentence for 19% of trials, because one wrong word makes a sentence wrong. Gating its text by agreement with the neural evidence kept 96% of its speed and reduced wrong commits to 4%. The gate needed only the stream, the published text and a light reference decoder, not the vendor's weights. This is the use case a vendor-neutral stream format is meant to enable.

Decoder adaptation set the speed. On sEMG, a decoder trained with half a session of the user's own data gave 71% higher eWPM than one trained on another day (8.02 vs 4.68). The trust layer made both safe, but only the in-session data made the system fast. A session-start block could serve both decoder adaptation and threshold calibration, but our results show that a threshold calibrated early in a day can be too permissive later the same day, and CRC needs at least 19 labelled items to commit at all. Such a block should therefore seed an online controller rather than fix the threshold.

How to state the guarantees. Our results suggest precise wording. With CRC: "if the calibration sentences and later sentences are exchangeable, then, averaged over the choice of the n calibration sentences and over a new sentence, the probability that the new sentence is committed wrongly is at most 0.05". A single session can exceed 0.05. With LTT: "under the same assumption, with probability at least 0.9 over the calibration sentences, the long-run wrong-commit rate at the chosen threshold is at most 0.05". With adaptive online re-calibration: "for any sequence of sentences, the average observed wrong-commit rate over the first T sentences is at most $0.05 + (1 - \lambda_1 + \eta)/(\eta T)$", which is 0.064 for $T = 816$ and $\eta = 0.05$. All three statements count only the errors that the system observes, so each should be reported with the measured silent-error rate, and none of them says anything about the error among commits.

What standardisation buys. One stream format let us apply one calibration, gate and risk controller to three device types with adapters of 148 to 568 lines of code. The format is deliberately small; its value is that a consumer such as the trust layer, or an independent auditor, can be written once.

9 Limitations

Validity of the evidence.

1. **One implant participant.** The only non-trivial test of the commit decision is Brain-to-Text '25, which contains one participant in a copy task, with online logits that we cannot re-run. Our intervals cover trials and days, not people.
2. **Composed sEMG sentences.** The Toronto data has isolated key presses only. Sentences are composed with oracle timing and no inter-key context, from a limited pool of presses that recur across sentences, and the test sentences come from the same corpus as the LM. Natural typing is untested, and the layer committed only 0–4% of sentences.
3. **MEG is a negative control.** The true sentence is in the 10-best list for 1.6% of MEG test instances, the layer commits nothing, and the 17 unique test sentences give few clusters. Our MEG decoder is weak, uses key times as training labels and sentence bounds from the keyboard log, and is not comparable head-to-head with published decoders.
4. **No live users.** All eWPM values come from a simulated user. Only two of its costs are matched to measurements of one gaze-typing user; the miss probability of a wrong committed word has no measurement; the model gives offers a more careful user than commits; and the populations of the MEG and sEMG data do not match the gaze-typing user. On MEG, two plausible settings reverse the comparison with gaze typing. A user study needs ethics approval.
5. **One seed, one split, simple LMs.** Every decoder and harness run used one seed, and the Brain-to-Text '25 and sEMG trust runs one fit/test split. The LMs are a character 6-gram and a word unigram trained on Tatoeba; 64 of the 1,628 Brain-to-Text '25 cue sentences occur in Tatoeba English, and near-duplicates of the sEMG test sentences were not removed from the LM text. **[TODO: R1: LM rebuilds without these sentences and re-runs; decoder seeds.]**
6. **Drift baselines reuse data.** The offline CRC baselines of the drift experiment were calibrated on data that also fitted the temperatures and the policy, and the trial order is taken from the file. **[TODO: R1: clean-calibration baseline.]**

Scope of the system.

7. **Silent errors are outside every guarantee.** With feedback on noticed errors only, `aci` gave 0.0515 (Brain-to-Text '25) and 0.0635 (sEMG).
8. **Guarantees are marginal or long-run.** CRC bounds the mean over calibration draws (41–45% of splits exceeded 0.05); online bounds are long-run averages; 50-trial windows reached 0.10–0.18. None bounds the error among commits.
9. **Stream-only gating cannot detect intent mismatch.** A real stream of another sentence was committed as often as matched input.
10. **Exact-match loss.** One wrong character makes a sentence wrong; a word- or meaning-tolerant loss could change the trade-offs. **[TODO: R1: sensitivity of the main results to a one-word-edit tolerance.]**

11. **The audit log is tamper-evident, not tamper-proof**; it has no signature, and in live use a wrong commit is known only after a correction.

10 Conclusion

We defined a minimal per-frame output format for neural text-entry decoders and a trust layer that reads only this format and decides whether to commit, offer or abstain under a stated risk bound. Re-fitted per device, the same implementation turned frequent wrong commits into rare ones on one intracortical participant at the speed of always committing and slightly faster than always offering, while one committed sentence in four was still wrong. On sEMG sentences composed from isolated presses it almost never committed, and on a weak MEG decoder it correctly never committed. Conformal risk control met its target on exchangeable splits, and online re-calibration kept the long-run rate near the target when the error level of the test data differed from the calibration data. The layer committed no text from signal-destroying streams, but it cannot detect a real stream of a different intended sentence, and it bounds the marginal risk, not the error among commits. All speed results come from a simulated user. The next steps are a study with gaze-typing users, which needs ethics approval; a controller for the error among commits; and a public benchmark built on the same harness and controls.

11 Ethics and data licences

This study uses only public, de-identified datasets. No new data were collected and no experiment with human participants was run, so no new ethics approval was required. The intracortical data come from a participant in the BrainGate2 pilot clinical trial [2]; we use only the publicly released copy-task logits and sentences. **[TODO: author decision: final wording of this statement and the venue's policy on secondary use of public clinical-trial data.]** We did not attempt re-identification. Raw recordings and participant data are not redistributed; our repository contains code, configurations, derived summary statistics and audit logs only. Table 6 lists every dataset and resource and how we use it.

Table 6: Datasets, resources and licences.

Dataset / resource	Licence	Use	Note
SpanishBCBL [5, 82]	CC BY-NC 4.0	MEG decoder training and evaluation	Research only; weights trained on it are not released for commercial use
Brain-to-Text '25 copy-task logits [2, 39]	CC0 1.0	intracortical trust evaluation	Companion code has no licence; we wrote our own loader
Toronto sEMG typing [8, 81]	CC BY 4.0	sEMG decoder and trust evaluation	Baseline code (CC BY-SA 4.0) not used
Tatoeba English / Spanish sentences	CC BY 2.0 FR (some CC0)	character LMs; composed sEMG sentences	Sentences from Tatoeba (`https://tatoeba.org`), by the Tatoeba contributors
CMUdict (commit 7479086)	BSD-2-Clause style	phoneme-to-word lexicon	Copyright notice kept
Brain2Qwerty code (Meta)	CC BY-NC 4.0	read only, not copied	Our decoders are our own implementations
Fidetype code and specification	Apache-2.0	all methods	—

12 Code and data availability

The specification (NeuralTextStream v0.1), the conformance suite with its golden vectors, the reference implementation (`fidetype`: adapters, decoders, trust layer, online risk control, audit log) and all experiment scripts and configurations are available under Apache-2.0 at `https://github.com/zhangzibo/fidetype`. The released code contains no model weights trained on Spanish-BCBL (CC BY-NC 4.0) and no copied Brain2Qwerty code. Every table and figure in this paper is generated from committed result files; the figure script is `papers/paper1/make_figures.py`, and the derived quantities of this revision (error among commits, 90th percentiles, pool risk) come from `papers/paper1/analysis_r1.py`. The datasets are available from their original sources (Table 6); each has a fetch note with a SHA-256 digest in the repository. **[TODO: tag the commit used for the paper and archive it (e.g. Zenodo DOI) before submission.]** Hardware: one AMD RX 7800 XT GPU (ROCm) for decoder training; all trust-layer runs use a CPU. **[TODO: report trust-layer run time per decision (only the audit-log cost is measured).]**

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A Notation

Table 7: Notation.

Symbol	Meaning	Symbol	Meaning
F_i	frames in the stream of item i	λ	commit threshold on s_i
V	vocabulary size of the stream	s_i	commit score of item i (Section 5.4)
K	size of the candidate list ($K = 10$)	n	number of calibration items
k	number of offered candidates	T	number of items in an online sequence
τ_f, τ_s	frame and sentence temperatures	η, γ	step sizes of the online controllers
c_i, f_i	commit loss, non-commit loss	α, δ	target risk, LTT failure probability

B User model parameters

Table 8 lists the parameters of the simulated gaze-typing user (word model: the user fixes single words). Only the fallback typing speed and the per-slot scan-and-select time are matched to measurements of one eye-gaze user with ALS [30]; all other values are assumptions. The baseline "gaze typing only" uses the fallback speed alone.

Table 8: Simulated-user parameters.

Parameter	Value	Basis
Fallback (gaze keyboard) speed	20 characters/min	measured, eye-gaze user LP1 [30]
Scan and select one of k word alternatives	$1.0 + 1.0 \log_2(k+1) + 1.5$ s (4.5 s for $k = 3$)	sum matched to 4.7 s [30]
Select one of k offered sentences	$1.0 + 0.5 \log_2(k+1)$ s	assumption
Reading time	0.3 s per word, 0.5 s per candidate	assumption
Navigate to and delete a wrong word	3.0 s (2 selections)	assumption
Miss probability of a wrong, unmarked word (commit, OFFER_WORDS)	0.05	assumption (no measurement)
Errors in offered candidates	always seen	assumption
Full retype overhead	1.5 s; chosen when cheaper than fixing words	assumption (rational user)
Abstain overhead	0.5 s, then fallback typing	assumption
Neural input speed, MEG / B2T '25 / sEMG	152 / 159.3 / 75.8 characters/min	device setting

C Per-device settings

Table 9 lists the parameters that the shared implementation fitted or used per device. The policy grid and all decoder settings are in the experiment configurations in the repository.

Table 9: Per-device settings of the shared trust-layer implementation. Fitted on each device's fit data (MEG: validation set). $\tau_s = 20.0$ is the upper end of the search interval. Max. guard risk: largest guard risk that still allows a commit.

	MEG	B2T '25, ours	B2T '25, published gate	sEMG (cal. / x-sess. / x-subj.)
Frame temperature τ_f	3.30	3.78	3.78	1.25 / 1.78 / 2.17
Sentence temperature τ_s	20.0	4.56	1.72	2.42 / 5.42 / 20.0
LM weight (beam)	0.5	1.0 (word LM; edit penalty 12, word bonus 3)	as ours	1.0 / 1.0 / 1.5
Candidate list K	10	10	published sentence + our 10 (bonus +2.0)	10
Plug-in commit threshold	0.5	0.5	0.7	0.7 / 0.3 / 0.3
Max. guard risk for commit	0.3	0.5	0.5	1.0 / 1.0 / 0.1
Offer size k	1	2	1	2 / 2 / 1
Fallback after offer	offer	offer	offer	offer
Guard	low-evidence threshold 0.5; consistency over the top 5; risk = max(low evidence, 1 − consistency), weights 1			
Adapter (lines of Python)	148	412	412	568

D MEG decoder, leakage checks and sensitivity

The asynchronous MEG decoder is a convolutional front end (320 channels) followed by 4 Conformer layers of width 256 [83], trained with CTC [40] and a keystroke auxiliary loss that uses key times as training labels (key pretraining, then CTC training on all 80 recording blocks). At inference, the forward pass takes the MEG segment, its length and the participant index; it has no key input. Table 10 compares the inference inputs with the published asynchronous decoder.

Table 10: Asynchronous setting: our decoder vs Brain2Qwerty v2.

Input at inference	Ours	v2 async [6]
MEG of the sentence segment	yes	yes
Segment start	first key − 0.4 s	first key − 400 ms
Segment end	last key press + 0.5 s	last key release + 400–500 ms
Per-key times or key count	no	no
Participant index	yes	yes
Key times as training labels	yes (auxiliary loss)	no

Leakage checks. The split is by a hash of the sentence text, and no test sentence occurs in the LM training text. A test sentence is "close" if a training sentence is within CER 0.25; 6 of the 17 test sentences are close. Without them, the CER was 0.519 (greedy) and 0.492 (beam).

Synchronous control decoder. A synchronous decoder with oracle key times (greedy CER 0.508, beam + LM 0.447; harness top-1 CER 0.451) gave the same trust results as the asynchronous decoder: always commit 2.714, trust 3.229 and gaze typing 3.155 eWPM, trust minus gaze [-0.02, +0.18] over sentences and [-0.07, +0.26] two-way. Over trials, this difference excluded 0 ([+0.03, +0.12]): trial-level intervals make the MEG result look like a gain, and sentence-level intervals do not. Both decoders take the sentence bounds from the keyboard log, so neither is a free-running decoder. **[TODO: R1: k-fold over sentence hashes to increase the number of unique MEG test sentences.]**

User-model sensitivity on MEG. Table 11 refits the policy under each user-model variant (asynchronous decoder). The trust layer equals always offering in every variant. The advantage over always commit depends on the miss probability, and the comparison with gaze typing

reverses with a faster fallback or slower selection. **[TODO: R1: the same grid on B2T '25 and sEMG, with the policy ranking and a misspecified-user evaluation.]**

Table 11: User-model sensitivity, MEG (asynchronous decoder; policy refitted per variant). eWPM; CIs over the 17 unique sentences and over the 19 participants.

Variant	Always commit	Trust	Gaze only	Trust − gaze [sent.]	{part.}
Main (miss 0.05, fallback 20 cpm)	2.58	3.20	3.15	[-0.04, +0.15]	{−0.07, +0.22}
Fallback 40 characters/min	4.31	5.15	6.31	[-1.29, -1.01]	{−1.31, −0.95}
Miss probability 0	3.24	3.20	3.15	[-0.04, +0.15]	{−0.08, +0.22}
Miss probability 0.2	1.24	3.20	3.15	[-0.04, +0.15]	{−0.08, +0.22}
Selection, scan and navigation ×2	2.54	3.00	3.15	[-0.23, -0.08]	{−0.26, −0.02}
No full retype	2.54	3.18	3.15	[-0.06, +0.13]	{−0.10, +0.21}

E Format details and audit log

Serialisations and conformance. The reference serialisations of NeuralTextStream are an NPZ file and a JSON Lines streaming form. A conformance suite runs 60 checks, each tied to a specification clause, and reports one of three cumulative levels: *core* (a consumer must accept the stream), *trust-ready* (modality, causality, calibration state and language are declared) and *provenance* (stream id, decoder name, version and weights hash). It ships 52 golden test vectors (32 NPZ, 20 JSON Lines); the reference consumer passes 52 of 52. On synthetic inputs, our MEG and sEMG adapters reach the provenance level and the intracortical adapter reaches trust-ready, because the published online decoder has no weights file to hash. Writing the suite found five bugs in our own producers and ten ambiguities in the specification text, now listed as open questions.

Audit log. Each decision can be appended to a JSON Lines log. Context entries are content-addressed (their id is the SHA-256 of the context) and hold the adapter, decoder (name, version, weights hash, training data and licence), LM, calibration (temperatures, risk-control method, α, δ, threshold, n) and policy versions; decision entries hold the action, scores, a hash of the stream and its frame range; correction entries link to the decision they correct. Each entry stores the hash of the previous entry and its own SHA-256 over the canonical JSON, so any edit, deletion or reordering breaks the chain; a separately stored head anchor detects truncation. The log stores no raw signals (numeric arrays longer than 64 values are rejected), stores text as a keyed hash by default, and pseudonymises subject and session ids. It is tamper-evident, not tamper-proof: there is no signature in v0.

Replay. We replayed the Brain-to-Text '25 test half (816 trials) with the log attached, for the plug-in gate and a CRC gate. The log held 1,723 entries and passed verification. It recorded the same wrong commits as the harness: 48 of 48 (plug-in) and 39 of 39 (CRC), of which 2 and 1 were silent. This is a consistency check of the replay: the corrections come from the known cue sentences, and in live use silent errors would be missing from the log. Every tampering case we tried was detected: editing an entry, editing with re-hashing, deleting, reordering, and cutting the tail (with the head anchor). The cost was about 0.1 ms and about 1.1 kB per decision. Figure 7 shows one entry of the audit report.

Wrong commit `dec-000041-ed60b4bd` (entry 44, hash `8cf2c6c67053b185...`)
Committed: "who you never expect to when"
Correction (known cue sentence, hash-linked): "who you never expect to win"; 2 character edits, 1 word edit
Scores: $\hat{p}_{\text{top1}}$ 0.5814, set size 3, guard risk 0.2857; reason: $\hat{p}_{\text{top1}} \geq 0.50$
Input: stream `1b4b60e8e4950710...`, frames [0, 335], item `b2t25-test-trial-196`
Decoder: published online RNN + our phoneme-to-word decoder; LM: Tatoeba character 6-gram + word unigram + CMUdict; adapter `fidetype.adapters.b2t25`
Calibration `cal-692fb52a73417552`: τ_f 3.7809, τ_s 4.5551, fit set: first half of each day (812 trials)
Policy `pol-2eaa4ff51919527a`: commit threshold 0.5, maximum guard risk 0.5, offer size 2

Figure 7: One entry of the audit report (B2T '25 replay, plug-in gate; public CC0 cue sentence). Each wrong commit is traced to the exact stream segment and to the decoder, LM, calibration and policy versions that produced it. Hashes are shortened.

Table 12: Online re-calibration methods for the commit threshold. L_t: realised loss of item t; T: number of items in the sequence; d: feedback delay in items. The two `aci` variants bound the long-run average of the *observed* losses for any sequence of items, without exchangeability; `window` and `decay` have no guarantee under shift.

Method	Update	Guarantee	Feedback
`window`	CRC on the last $N = 200$ items with known truth	none under shift[a]	full
`decay`	weighted CRC on all past items, weights ρ^{age}, $\rho = 0.99$	none under shift[a]	full
`aci`	$\lambda_{t+1} = \max(0, \lambda_t + \eta(L_t - \alpha))$, $\eta = 0.05$	$\frac{1}{T}\sum_t L_t \leq \alpha + \frac{\lambda_{T+1} - \lambda_1}{\eta T}$ [b]	realised only
`aci_window`	window CRC at level α_t; $\alpha_{t+1} = \min(2\alpha, \alpha_t + \gamma(\alpha - L_t))$, $\gamma = 0.005$	$\frac{1}{T}\sum_t L_t \leq \alpha + \frac{\alpha_1 - \alpha_{T+1}}{\gamma T}$ [c]	full + realised

[a] The weighted-CRC bound $\alpha + \sum_i \tilde{w}_i \, d_{\text{TV}}(Z, Z^{(i)})$ [58, 59] cannot be computed. Items whose truth is unknown are left out of the window, which can bias it.
[b] The right-hand side is a certificate computed after the fact. It is bounded in advance only if λ_t stays bounded, which holds for scores ≤ 1 and $f \equiv 0$; then $\frac{1}{T}\sum_t L_t \leq \alpha + \frac{1 - \lambda_1 + \eta(1+d)}{\eta T}$, which is 0.064 for $T = 816$, $\eta = 0.05$, $d = 0$ and the B2T '25 start threshold.
[c] The upper clip at 2α only lowers α_t, so the bound stays valid. α_t stays bounded below because below the CRC feasibility level ($\approx 1/(N+1)$) nothing is committed and α_t rises.
All bounds count only the losses that the controller observes. Items whose feedback has not arrived are skipped; in our runs all feedback arrived after the delay, except in the "noticed errors only" runs, where a silent error is fed back as a correct commit.

F Online re-calibration methods

G Additional risk-control results

Figure 8 shows the spread of the test wrong-commit rate over the 200 random calibration/test splits for 100 and 408 calibration items. For `aci` on Brain-to-Text '25 (our decoder), step sizes η of 0.01, 0.02, 0.05, 0.1 and 0.2 gave wrong-commit rates of 0.0527, 0.0515, 0.0502, 0.0502 and 0.0502, with a-priori bounds of 0.114, 0.083, 0.064, 0.058 and 0.054. On sEMG session-calibrated, pure offline CRC was below the target (0.0417), and the online methods gave 0.0454–0.0490; on sEMG cross-subject nothing could be committed.

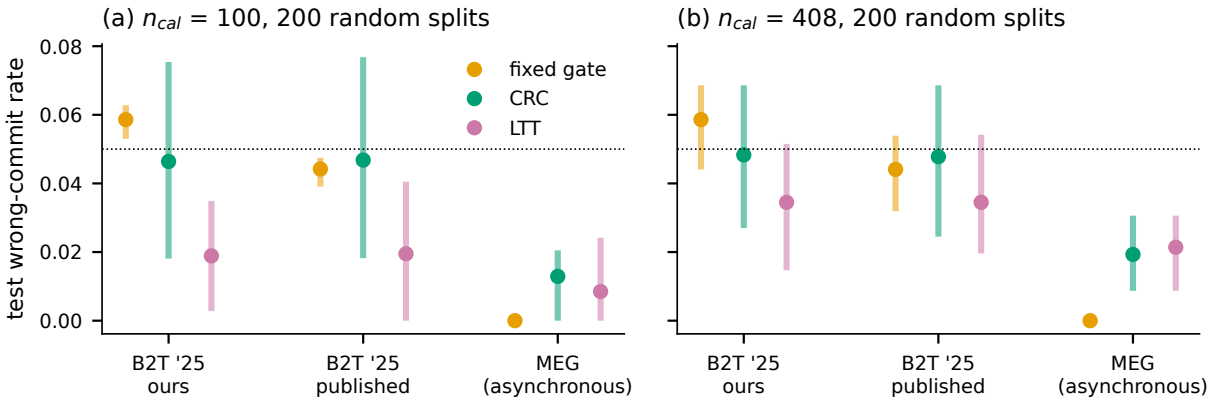


Figure 8: CRC bounds the mean over splits; LTT bounds the calibration-conditional risk with probability $1 - \delta$**.** Test wrong-commit rate over 200 random calibration/test splits with (a) 100 and (b) 408 calibration items. Dots: mean; bars: 5th to 90th percentile (the 90th percentile matches $\delta = 0.1$). Dotted line: $\alpha = 0.05$. A test rate above α in a single split is not a violation of either guarantee.